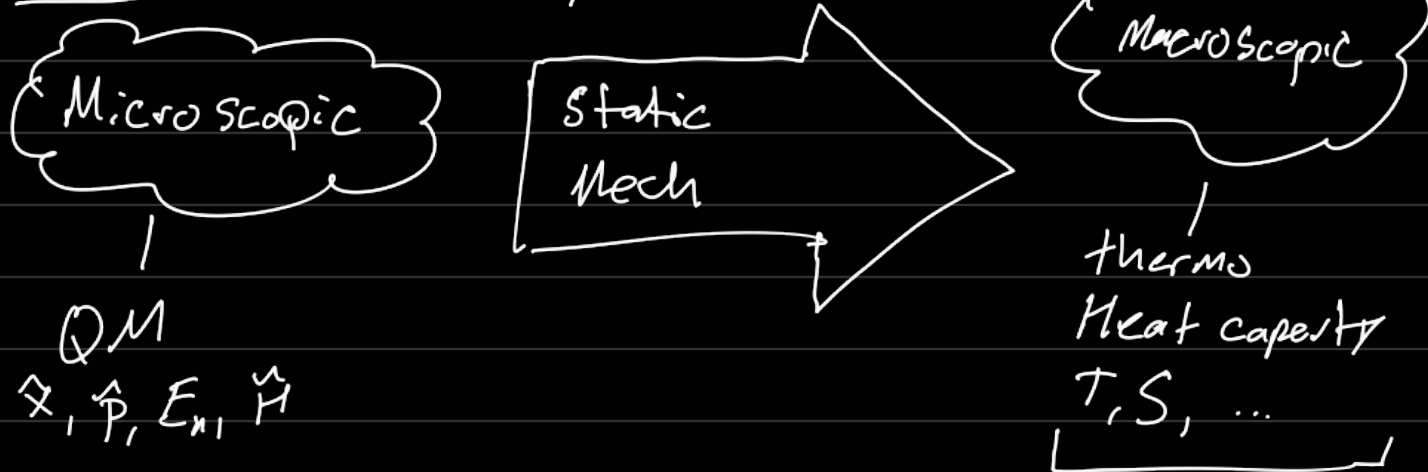


- 5.2.3, 5.2.5, 5.2.6, 5.3.2



Microstate: $\hat{H}|\psi\rangle = E|\psi\rangle$

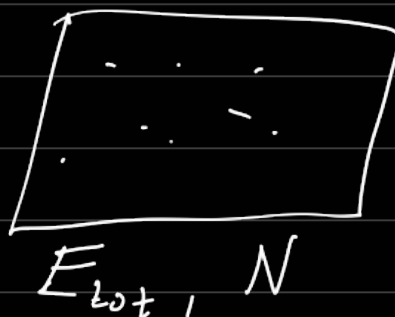
• Ex: $E_n \sim \frac{1}{n^2}$

* Systems:

• Microcanonical Ensemble:

• $\frac{\partial E}{\partial t} = \frac{\partial N}{\partial t} = 0$

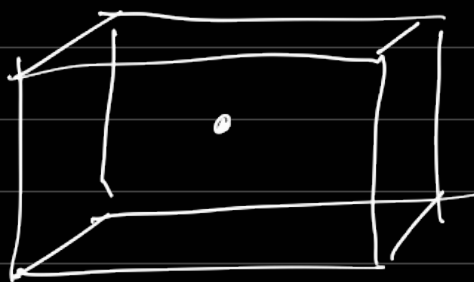
E, N : constant



• Canonical: $\frac{\partial E}{\partial t} \neq 0$ E doesn't have to be
 $\frac{\partial N}{\partial t} = 0$ N : const

Grand-can: Both E, N can vary

* Microstates: $W(E) = \#$ of microstates at Energy E



$\hat{H}|\psi\rangle = E|\psi\rangle, \quad H = p^2/2m$

$E_n = \frac{\hbar^2 v^2}{L^2} (\underbrace{n_x^2}_{\sim} + \underbrace{n_y^2}_{\sim} + \underbrace{n_z^2}_{\sim})$

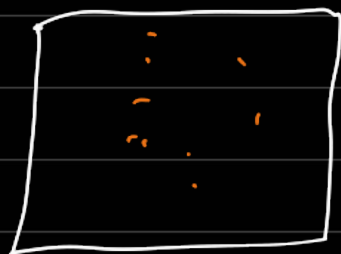
$$n_2 = 5, n_x = n_y = 0$$

$$n_x = 2, n_y = 3, n_z = 0$$

$$2^2 + 3^2 = 25$$

$$E_s \approx 25$$

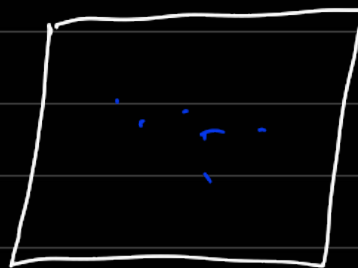
Counting:



System 1

E_1, N_1

• $W(E_1)$



2

E_2, N_2

• $W(E_2)$

$$W(E_1, E_2) = W(E_1) W(E_2)$$

* Fundamental assumption: All microstates have eq. prob. at equilibrium

ex: $W(E_1) : p(E_i) = \frac{1}{W(E_1)}$

• Stat. independence:

prob. multiply

$$p(E_1, E_2) = p(E_1) p(E_2) = \frac{1}{W(E_1)} \cdot \frac{1}{W(E_2)}$$

↓

$$p(E_1, E_2) = \frac{1}{W(E_1, E_2)} \leftarrow \text{total \# of states}$$

Total Microstates for (1 + 2)

$$W(E_1, E_2) = W(E_1) W(E_2)$$

System 1

System 2

$$\{1, 2\}$$

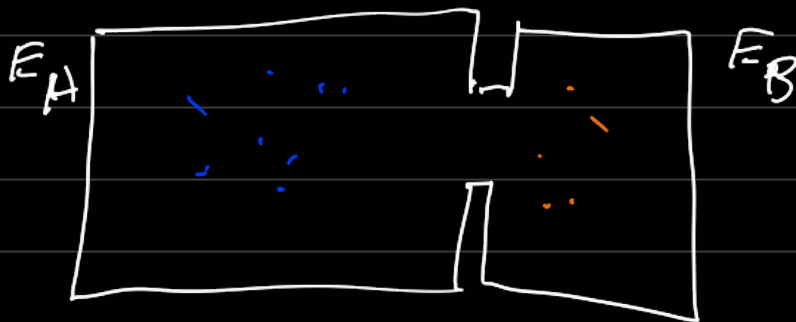
$$(2)$$

$$\{1, 2, 3\}$$

$$(3)$$

Total: $\{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3)\}$ $\frac{2 \cdot 3}{2 \cdot 3} = 6$

- Two system: let them interact



- $W(E_1, E_2) = W(E_1)W(E_2)$ (Before)
- Idea: In equilibrium the system will tend to the maximum of $W(E_1, E_2)$

- $dE = dE_A = -dE_B$

- E_2 will give energy to E_1

$$\frac{\partial}{\partial E} (W_A W_B) = 0$$

$$\frac{\partial W_A}{\partial E} W_B + \frac{\partial W_B}{\partial E} W_A = 0 \quad \leftarrow$$

Chain rule: $\frac{\partial W_A}{\partial E} = \frac{\partial E_A}{\partial E} \frac{\partial W_A}{\partial E_A}$

$$\left(\frac{\partial \ln(W_A)}{\partial E_A} \right)_V = \left(\frac{\partial \ln(W_B)}{\partial E_B} \right)_V = \text{const.}$$

$$= \frac{1}{T k_B} = \left(\frac{\partial S}{\partial E} \right) = \frac{1}{T}$$

$$S = k_B \ln(W)$$

• ... $\frac{1}{T}$

S: Counting Microstates

• mol $\sim 10^{23}$ particles

$\rightarrow$ Spin $\uparrow$ or $\downarrow$

(1) $\uparrow \downarrow \uparrow \downarrow \downarrow \dots$

(2) $\uparrow \uparrow \uparrow \downarrow \downarrow \dots$

every possible combination

$$W(\text{spin}) = 2^{(10^{23})}$$

• 10^{23} : in meters

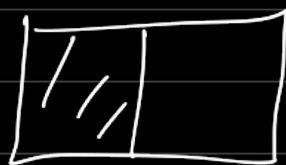
(a) 10^{16} m: closest star to us

(b) 10^{22} m: distance to the Andromeda gal.

$$S = k_B \ln(W) = 10^{23} k_B \ln(2) \quad N$$

$S \sim N$

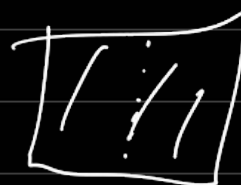
5.2.5



V

$$P_i = \frac{1}{W_i}$$

Single E.



$2V$

$$P_f = \frac{1}{W_f}$$

$$W_f \sim 2W_i$$

$N = N_A$: av. # $\sim 10^{23}$

• Entropy: $\hat{P}_i = \left(\frac{1}{W_i} \right)^{N_A} = \underbrace{\frac{1}{W_i}}_1 \underbrace{\frac{1}{W_i}}_2 \dots \underbrace{\frac{1}{W_i}}_{N_A}$

$$W_f = 2W_i$$

$$\hat{P}_f = \left(\frac{1}{2W_f} \right)^{N_A}$$

$$\hat{P}_i = \left(\frac{1}{2} \right)^{N_A}$$

$$\frac{1}{\beta_f} = \left[\frac{1}{2} \right]$$

$$\Delta S = N_A k_B \ln(2)$$

5.2.6

* Recall $W(E_A, E_B) = W_A W_B$ (Before)

over time $W(E_A, E_B)$: Systems in equilibrium

• At eq. W : $\frac{dW_{AB}}{dE} \bigg|_{E^*} = 0$

* $\tilde{W}(E_A, E_B) \approx W(E^*)$

$$d\tilde{W}(E_A, E_B) \geq d(W(E_A, E_B)) \geq 0$$

• $d(W_A W_B) \geq 0$

$$(dW_A) W_B + W_A dW_B \geq 0$$

• other assumption $dE_B = dE_A = dE$

$$\frac{dW_A}{W_A} + \frac{dW_B}{W_B} \geq 0$$

$$\frac{d \ln(W_A)}{dE} + \frac{d \ln(W_B)}{dE} \geq 0$$

$$\frac{dE_B}{dE} = 1$$

• chain rule: $\frac{d \ln(W_A)}{dE} = \frac{\partial E_A}{\partial E} \frac{\partial \ln(W_A)}{\partial E_A}$

$$\left(\frac{\partial E_A}{\partial E} \right) \left(\frac{\partial \ln(W_A)}{\partial E_A} \right) + \left(\frac{\partial E_B}{\partial E} \right) \left(\frac{\partial \ln(W_B)}{\partial E_B} \right) \geq 0$$

$$\uparrow$$

1

$$\uparrow$$

-1

Instead of $\frac{dE}{\partial E_A} \frac{\partial \ln(W_B)}{\partial E_B} = \frac{\partial E_A}{\partial E_B} \frac{\partial \ln(W_B)}{\partial E_B}$

Plug in:

$$\left(\underbrace{\left(\frac{\partial \ln(W_A)}{\partial E_A} \right)}_{\frac{1}{k_B T_A}} - \underbrace{\left(\frac{\partial \ln(W_B)}{\partial E_B} \right)}_{\frac{1}{k_B T_B}} \right) \geq 0$$

$$\left(\frac{1}{k_B T_A} - \frac{1}{k_B T_B} \right) \geq 0$$

$$\frac{1}{T_A} \geq \frac{1}{T_B} \Rightarrow$$

$$\boxed{T_B \geq T_A}$$

$$* \quad \frac{1}{T} = k_B \left(\frac{\partial S}{\partial E} \right) \quad S \sim N$$

